Time Series Homework 3

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February 1, 2026

1 Task 2

1.1 Data Transformations

Four series were constructed for comparative analysis:

1. Original GDP per capita series
2. Natural logarithm of GDP per capita
3. First differences of original series
4. First differences of logarithmic series

1.2 Stationarity Testing

Three formal tests were employed with their respective null hypotheses:

- **Augmented Dickey-Fuller (ADF)**: H_0: Unit root (non-stationary)
- **KPSS**: H_0: Stationarity around a constant
- **Phillips-Perron (PP)**: H_0: Unit root

For ADF testing, lag length was selected automatically using Akaike Information Criterion (AIC).

1.3 Model Selection and Estimation

The AutoRegressive (AR) model of order p was specified as:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where y_t is the stationary series, c is a constant, ϕ_i are autoregressive coefficients, and ε_t is white noise.

Model order selection used multiple criteria:

- Akaike Information Criterion (AIC)
- Bayesian Information Criterion (BIC)
- Hannan-Quinn Information Criterion (HQIC)
- Partial Autocorrelation Function (PACF) analysis

1.4 Stationarity Analysis

Table 1: Stationarity Test Results

Series	**ADF p-value**	**ADF Stat.**	**KPSS p-value**	**KPSS Stat.**	**PP p-value**
Original GDP	0.9989	2.2736	0.0100	1.7191	1.0000
Log GDP	0.9873	0.5860	0.0100	2.1399	0.9803
First Differences	7.35e-06	-5.2380	0.0100	1.2552	3.28e-16
First Diff of Logs	**8.21e-07**	**-5.6877**	**0.1000**	**0.1251**	**4.77e-25**

The results in Table 1 indicate that only the first differences of logarithmic GDP per capita are stationary according to both ADF (rejects unit root) and KPSS (fails to reject stationarity) tests.

Figure 1: Time Series Plots of All Transformations

Figure 1 shows that while the original and log series exhibit clear upward trends, the differenced series fluctuate around zero, consistent with stationarity.

1.5 Autocorrelation Analysis

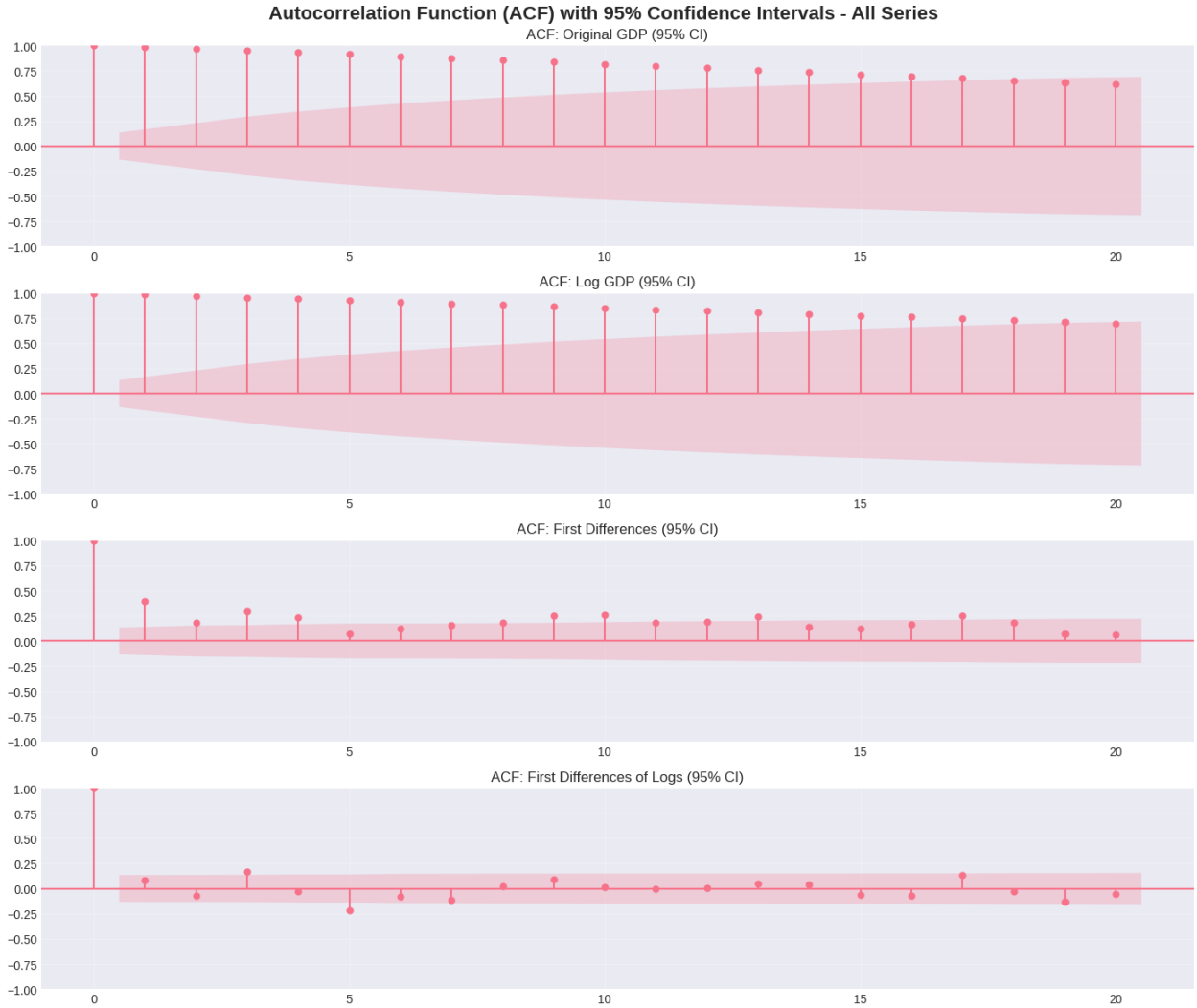

Figure 2: ACF Plots of All Series

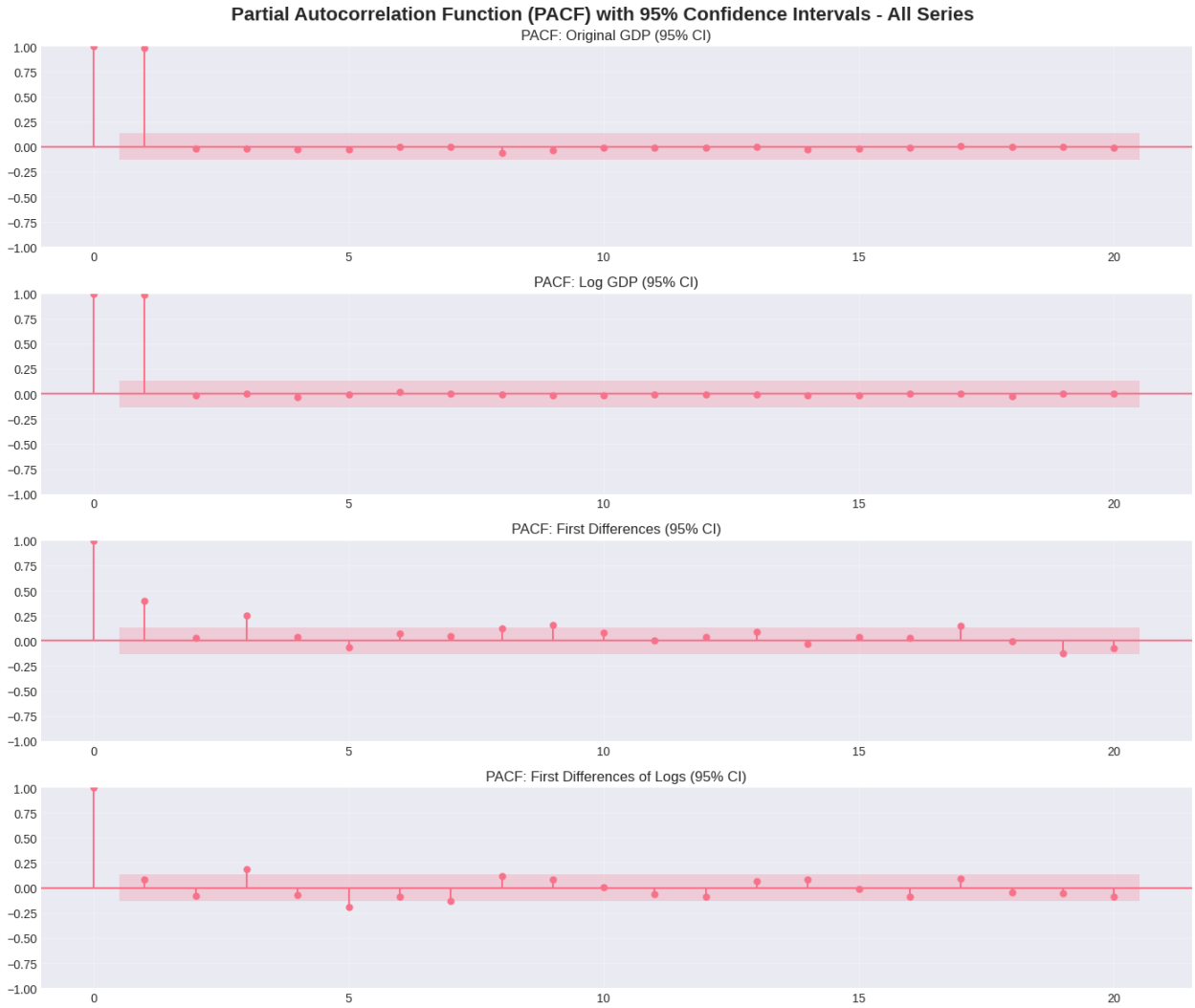

Figure 3: PACF Plots of All Series

The ACF and PACF plots reveal:

- Slow decay in ACF for original and log series (non-stationarity)
- Quick cutoff in ACF for differenced series (stationarity)
- Significant PACF at lag 1 and 3 for differenced log series

1.6 AR Model Selection

Table 2: Information Criteria for AR Order Selection

AR Order (p)	AIC	BIC	HQIC
0	**-614.601**	**-607.859**	**-611.877**
1	-610.227	-600.129	-606.146
2	-608.356	-595.002	-603.018
3	-608.843	-592.232	-602.248
4	-607.744	-587.876	-599.892
5	-607.585	-584.460	-598.475

All information criteria (AIC, BIC, HQIC) unanimously select AR(0) as the optimal model order (Table 2).

1.7 AR(0) Model Estimation

The selected AR(0) model for GDP growth rates is specified as:

$$\Delta \ln(y_t) = 0.0149 + \varepsilon_t$$

Table 3: AR(0) Model Estimation Results for French GDP Growth Rates

Coefficient	Estimate	Std. Error	z-statistic	p-value
Constant (c)	0.0149	0.0039	3.802	0.0001

1.8 Model Diagnostics

Table 4: Diagnostic Test Results for AR(0) Model Residuals

Test	Result
Ljung-Box Q(10) Test	
Test Statistic	26.5852
p-value	0.0030
Conclusion	Significant autocorrelation present
Jarque-Bera Normality Test	
Test Statistic	1196.2121
p-value	¡ 0.0001
Conclusion	Residuals not normally distributed

The diagnostic tests (Table 4) reveal important limitations of the AR(0) model:

1. **Autocorrelation**: The Ljung-Box test strongly rejects the null hypothesis of no autocorrelation (p = 0.0030), indicating significant serial correlation in the residuals.

2. **Normality**: The Jarque-Bera test overwhelmingly rejects normality (p ¡ 0.0001), suggesting residuals have fat tails or other non-normal features.

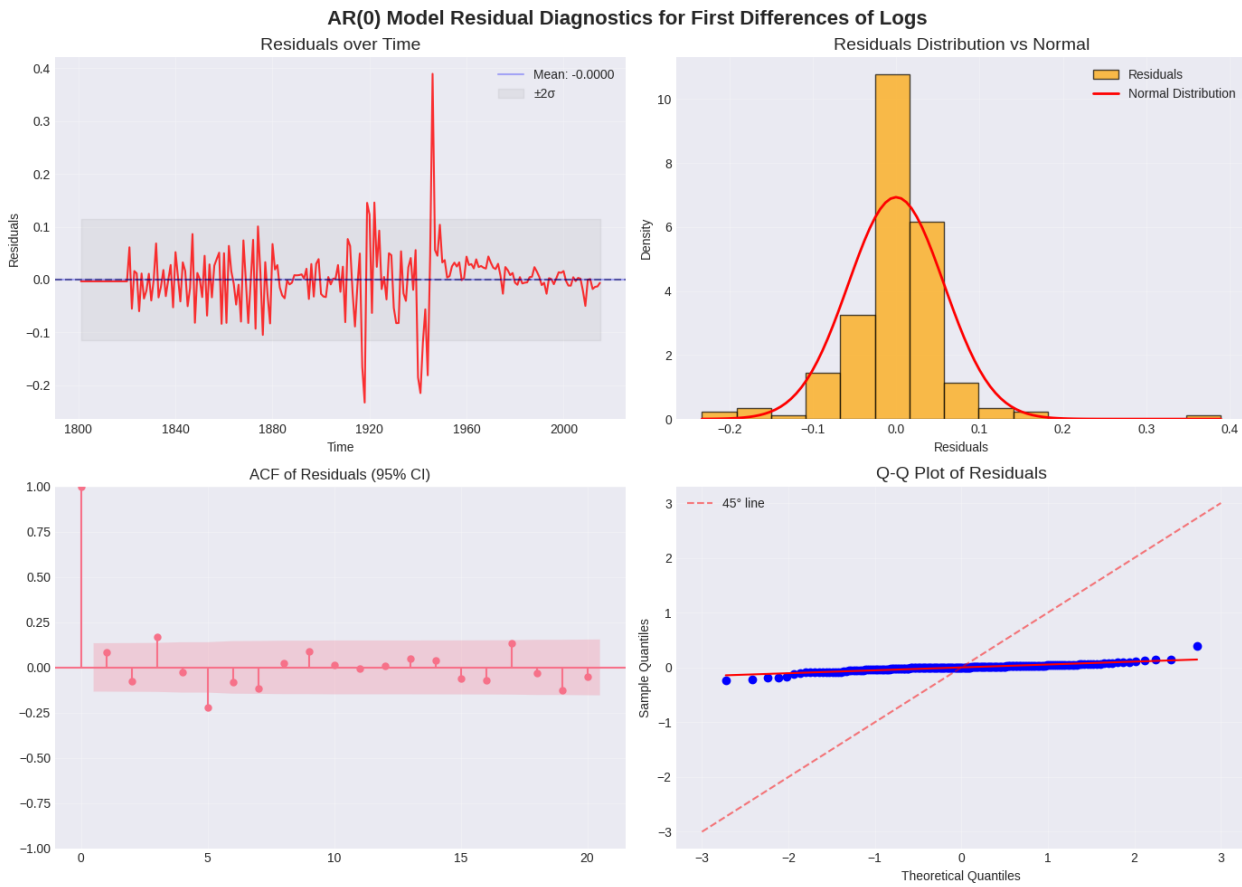

Figure 4: Residual Diagnostics for AR(0) Model

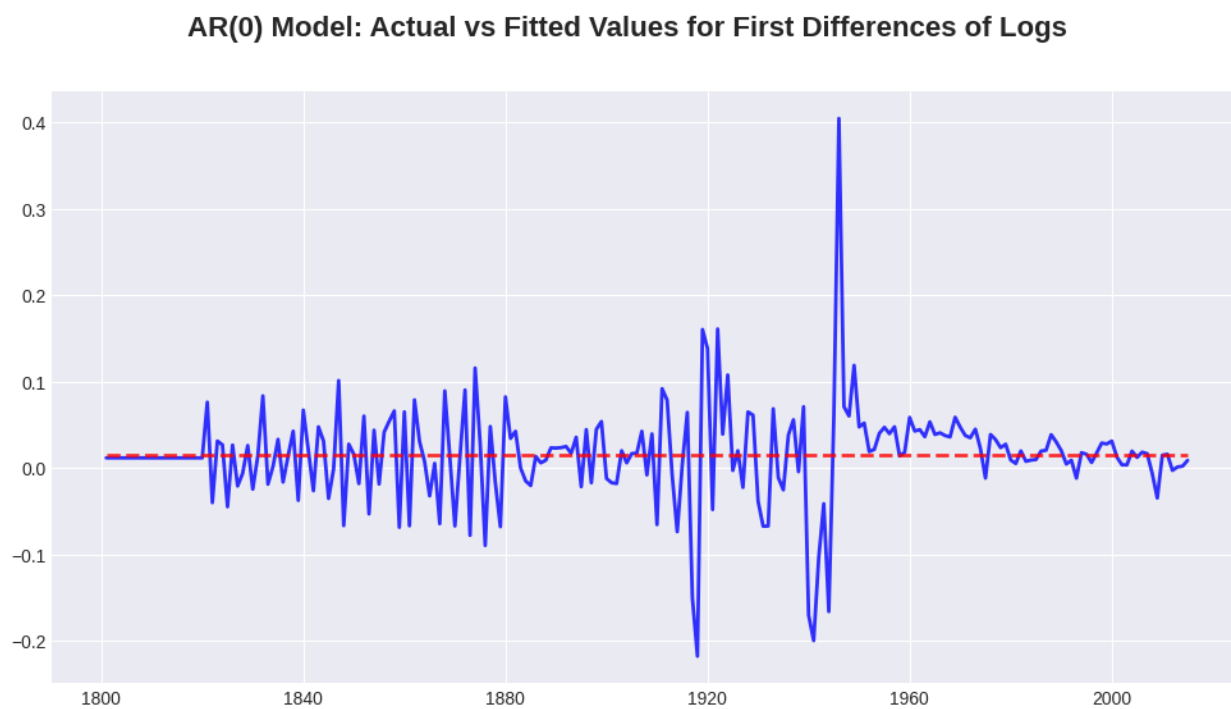

Figure 5: Actual vs Fitted

Results show that AR(0) is unbiased, but this is a poor model due to unexplained volatility, autocorrelation in residuals and non-normality. Dynamics of GDP are explained by external shocks without autoregression.

2 Task 3

2.1 Data Selection and Transformation

For Task 3, the Dow Jones Inflation-Adjusted Industrial Average (REALDJIA) was selected from the DJA dataset. The series contains 1,431 monthly observations. Two series were analyzed:

1. Original REALDJIA series
2. First differences of REALDJIA series

2.2 Stationarity Testing

The same three formal tests were applied to assess stationarity:

- **Augmented Dickey-Fuller (ADF)**: Lag length selected using BIC criterion

- **KPSS**: Null hypothesis of stationarity around a constant
- **Phillips-Perron (PP)**: Non-parametric test for unit root

2.3 Stationarity Analysis

Table 5: Stationarity Test Results for REALDJIA

Series	ADF p-value	ADF Stat.	KPSS p-value	KPSS Stat.	PP p-value
Original REALDJIA	0.8224	-0.7888	0.0100	4.4341	0.7576
First Differences	**0.0000**	**-35.6962**	**0.1000**	**0.0463**	**0.0000**

Table 5 presents clear evidence that the original REALDJIA series is non-stationary, while its first differences are stationary according to all three tests. The ADF and PP tests reject the unit root hypothesis for first differences (p ¡ 0.01), while the KPSS test fails to reject stationarity (p = 0.10).

2.4 Autocorrelation Analysis

Figure 6: Time Series Plots: Original REALDJIA and First Differences

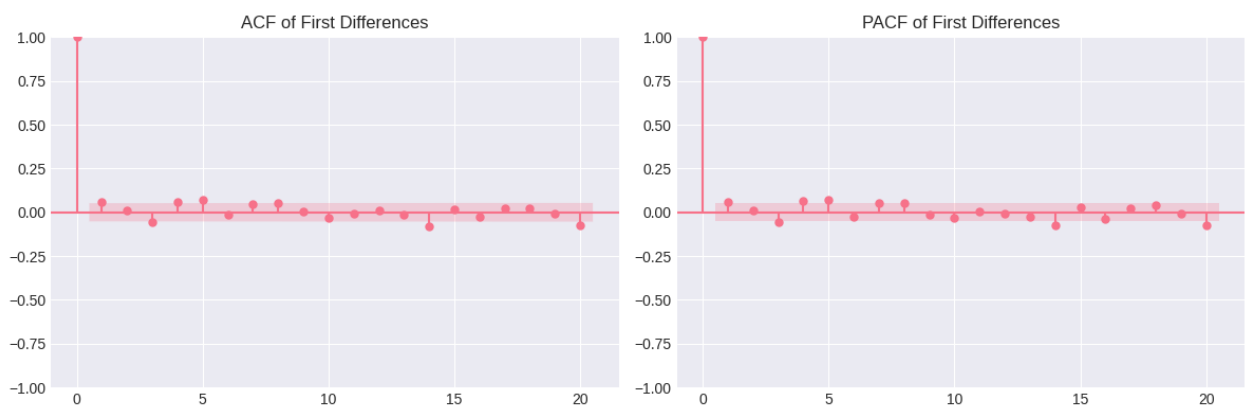

Figure 7: ACF and PACF for First Differences of REALDJIA

2.5 ARIMA Model Selection

Since first differences are stationary ($d = 1$), ARIMA models with various $(p, 1, q)$ specifications were estimated and compared using information criteria:

Table 6: Information Criteria for ARIMA Model Selection (d=1)

Model (p,1,q)	**AIC**	**BIC**	**Log Likelihood**
ARIMA(1,1,0)	-4292.77	-4282.23	2148.38
ARIMA(0,1,1)	-4292.62	-4282.09	2148.31
ARIMA(1,1,1)	-4290.77	-4274.97	2148.38
ARIMA(2,1,0)	-4290.91	-4275.11	2148.45
ARIMA(0,1,2)	-4291.17	-4275.37	2148.58
ARIMA(2,1,2)	**-4308.92**	**-4282.59**	**2159.46**

The ARIMA(2,1,2) model demonstrates the lowest AIC (-4308.92) and BIC (-4282.59) values (Table 6), indicating it provides the best balance between model fit and complexity.

2.6 ARIMA(2,1,2) Model Estimation

The selected ARIMA(2,1,2) model for REALDJIA can be expressed as:

$$\Delta y_t = -0.4198\Delta y_{t-1} - 0.7875\Delta y_{t-2} + \varepsilon_t + 0.4982\varepsilon_{t-1} + 0.8379\varepsilon_{t-2}$$

with $\varepsilon_t \sim \mathrm{N}(0, 0.0029)$.

Table 7: ARIMA(2,1,2) Model Estimation Results for REALDJIA

Coefficient	**Estimate**	**Std. Error**	**z-statistic**	**p-value**
AR(1): ϕ_1	-0.4198	0.0610	-6.880	0.0000
AR(2): ϕ_2	-0.7875	0.0480	-16.421	0.0000
MA(1): θ_1	0.4982	0.0569	8.763	0.0000
MA(2): θ_2	0.8379	0.0448	18.712	0.0000
σ^2	0.0029	6.3e-05	45.333	0.0000

All coefficients in the ARIMA(2,1,2) model (Table 7) are statistically significant (p ¡ 0.001). The negative AR coefficients suggest mean-reverting behavior, while positive MA coefficients indicate persistence of shocks.

2.7 Model Diagnostics

Table 8: Diagnostic Test Results for ARIMA(2,1,2) Model Residuals

Test	**Result**
Ljung-Box Q(10) Test	
Test Statistic	0.4569
p-value	1.0000
Conclusion	No significant autocorrelation
Jarque-Bera Normality Test	
Test Statistic	1298.53
p-value	0.0000
Conclusion	Residuals not normally distributed
Heteroskedasticity Test	
Test Statistic	0.42
p-value	0.0000
Conclusion	Significant heteroskedasticity

The diagnostic tests (Table 8) reveal:

1. **Autocorrelation**: The Ljung-Box test (p = 1.0000) indicates no significant serial correlation in residuals, confirming adequate model specification.

2. **Normality**: The Jarque-Bera test rejects normality (p ¡ 0.0001), consistent with financial time series that often exhibit fat tails.

3. **Heteroskedasticity**: Significant heteroskedasticity is present (p ¡ 0.001), suggesting volatility clustering characteristic of financial returns.

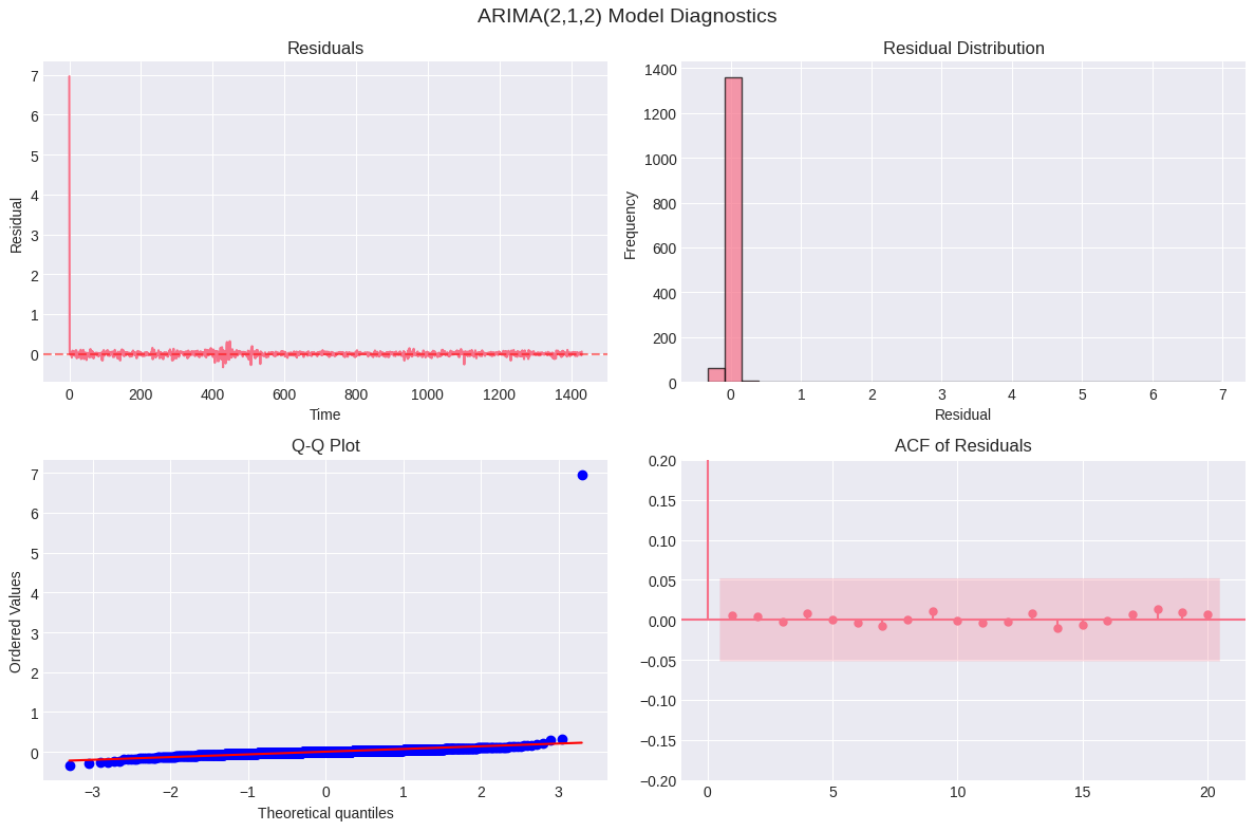

Figure 8: Residual Diagnostics for ARIMA(2,1,2) Model

Figure 8 shows the residuals appear random with no discernible pattern, supporting the adequacy of the ARIMA(2,1,2) specification. The Q-Q plot confirms deviation from normality, particularly in the tails.

2.8 Conclusion

The analysis demonstrates that REALDJIA is integrated of order 1 (I(1)), requiring first differencing to achieve stationarity. The ARIMA(2,1,2) model provides the best fit among considered specifications, capturing both autoregressive and moving average dynamics in the differenced series. While the model successfully removes serial correlation from residuals, the non-normality and heteroskedasticity suggest potential for improvement.